# Phonon Pulse Templates for SuperCDMS SNOLAB Run 4

K-line event selection · free-pretrigger two-exponential fits · 1x1 and NxM (PCA) templates

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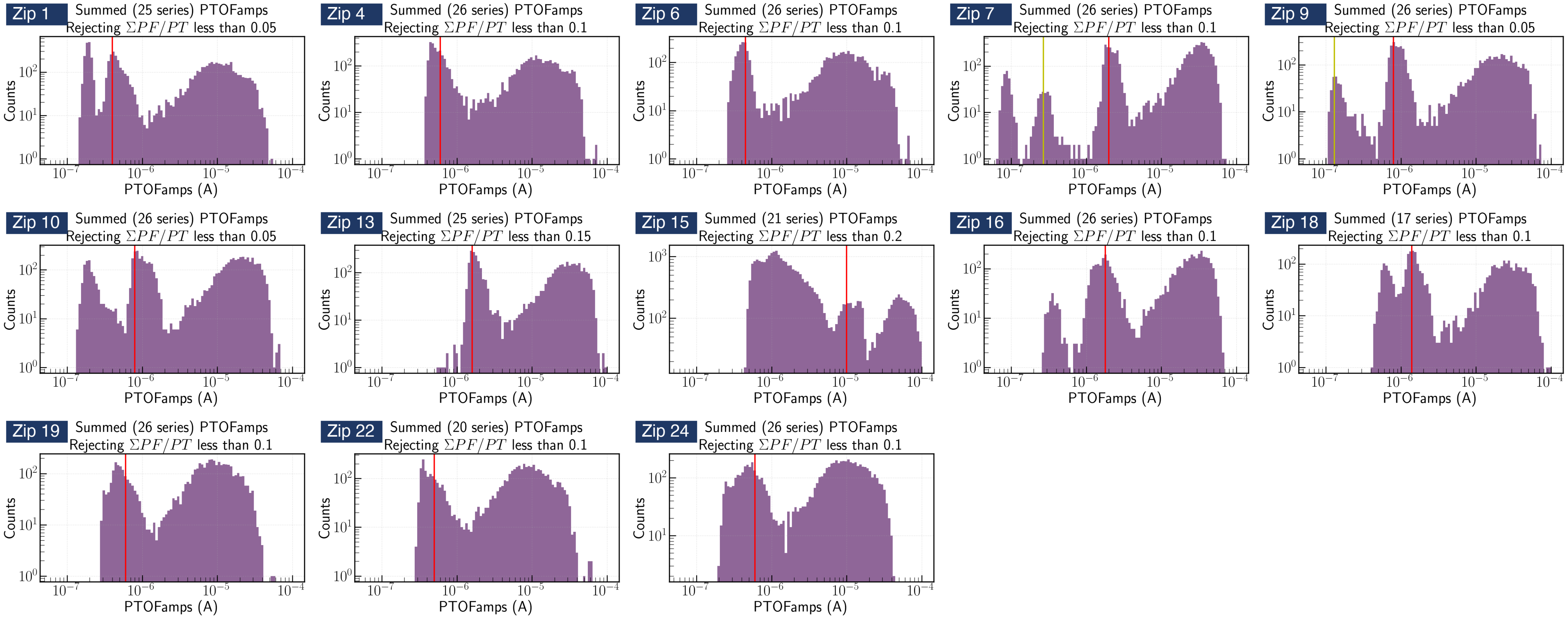
# From the Ge-activation K-line to an event sample

Goal: one clean, minimally-biased pulse population per detector × channel

- Per detector, the ops-shift study marked the 10.37 keV K-line position in PTOFamps — the red line in each panel below
- Our event selection: a PTOFamps window bracketing the red line (position ×/÷ 1.35) — deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata
- 13 detectors (zips) × 27–30 series → ~120 GB raw cache; every later cut stays explicit and reversible

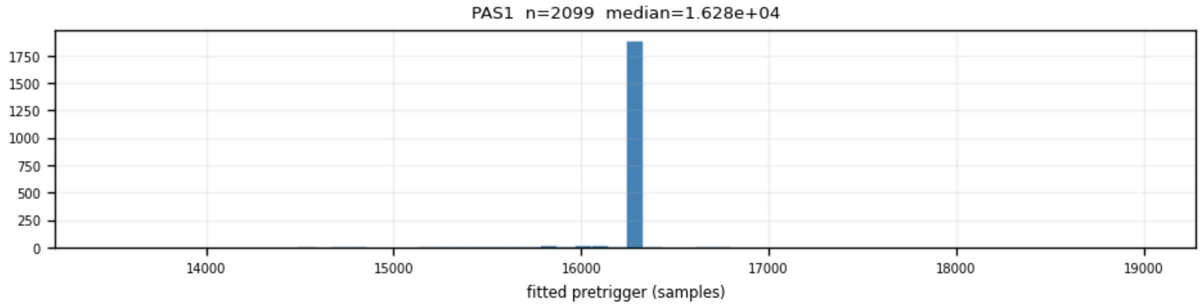



*All 13 detectors, PTOFamps spectra (ops-shift study): red line = fitted mean of the 10.37 keV K-line. Selection window per detector = red-line position ×/÷ 1.35.*

# Per-trace algorithm: low-pass → free-pretrigger fit → align

- 1. 100 kHz 4th-order Butterworth low-pass (steady-state init — no filter start-up transient)
- 2. Baseline = median of samples 2000–12000, subtracted; normalize by the global trace peak
- 3. Two-exponential fit — all 5 parameters free, including the onset:
  - $y(t) = A\cdot[\exp(-(t-t_0)/\tau\_fall) - \exp(-(t-t_0)/\tau\_rise)] + b$, $t_0 \in 16050 \pm 3000$ samples
  - fit window 12550–24050 (stride 4); $\tau\_rise \in [1\ \mu s, 5\ ms]$, $\tau\_fall \in [10\ \mu s, 20\ ms]$
- 4. Quality numbers — fit_ok := A>0 and 0<τ_rise<τ_fall; NRMSE := RMS(residual)/fit peak (recorded, no cut yet)
- 5. Align — shift the measured trace by (fitted onset − 16050), sub-sample interpolation: a pure translation

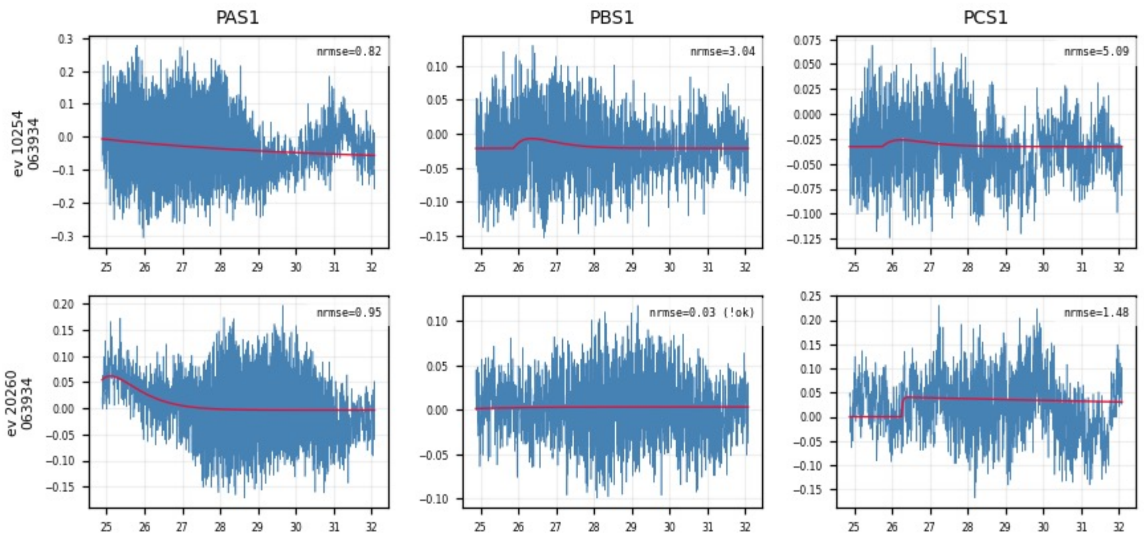
• Why a free onset? Real trigger times vary event-by-event — fitted onsets cluster ≈230 samples after the nominal 16050. Pinning the onset distorts every other parameter.



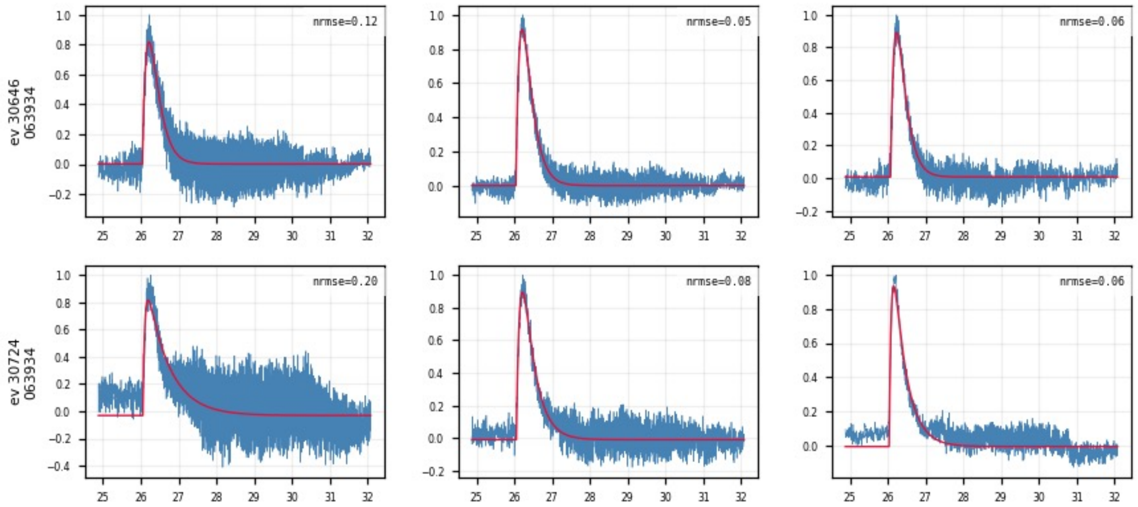

*Z7 PAS1: fitted onset distribution — offset from nominal, narrow core*

# Fit quality at a glance — event × channel grids

- One row per event, one column per channel: low-passed trace (blue) vs 2-exp fit (red), NRMSE stamped per panel
- The same event fits consistently across channels; a noise trigger fails in all channels at once
- Every figure carries its full processing chain in the header → self-documenting



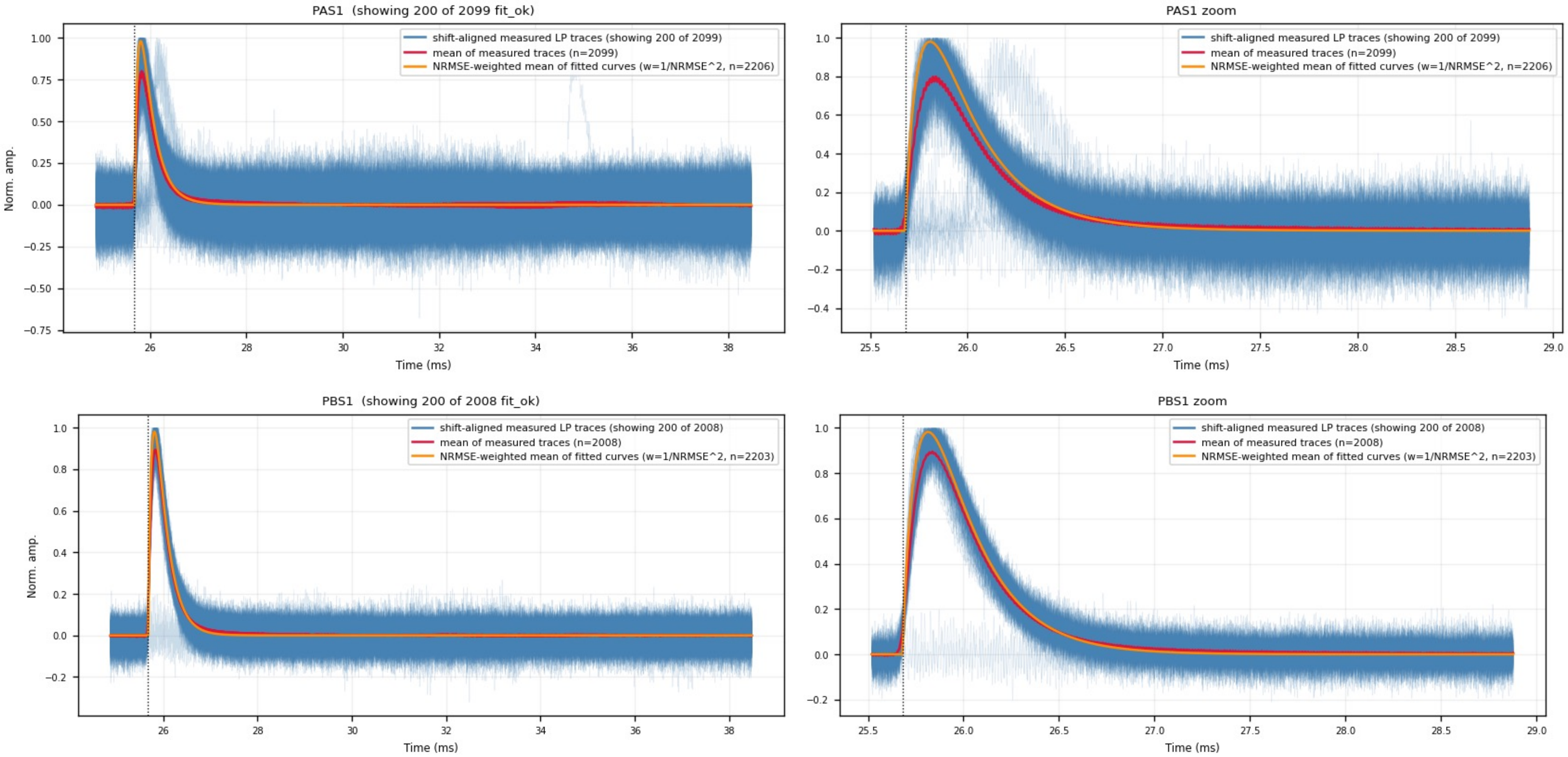

Noise triggers: the fit fails in every channel at once




K-line events: consistent good fits across channels

# Alignment result — overlaid measured traces

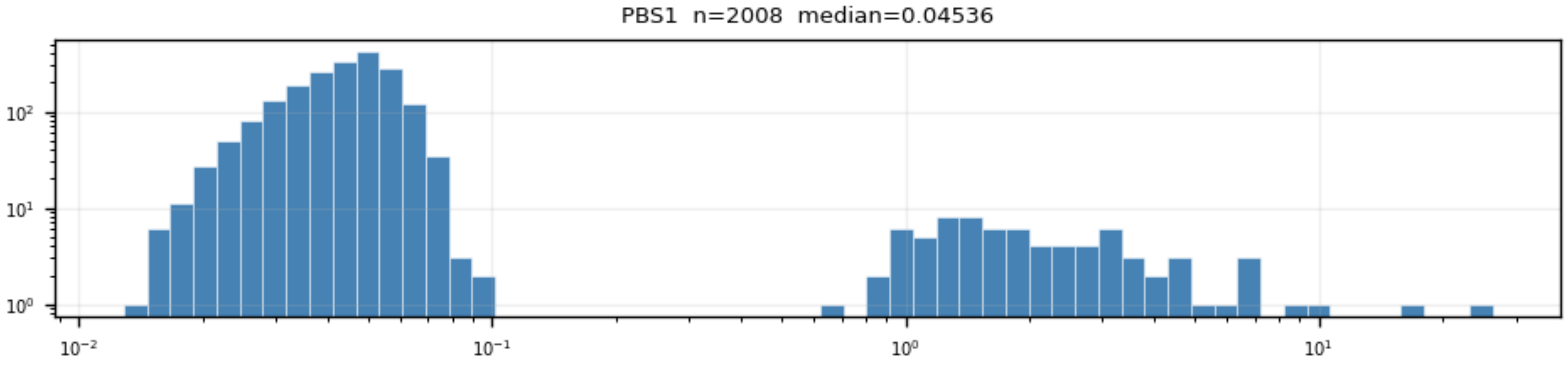
- All fit_ok traces shifted to the common onset (blue) + point-by-point mean (red) + NRMSE-weighted mean of the fitted curves (orange, $w = 1/\max(\mathrm{NRMSE}, 0.01)^2$)
- Quiet detectors give a tight bundle — the template input is well defined; right panels zoom on the peak



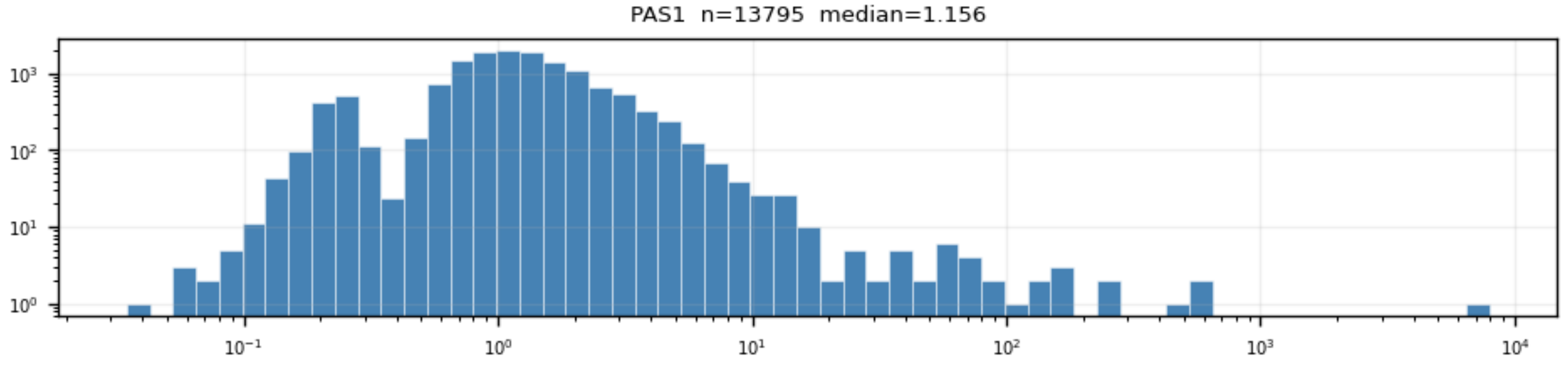

*Z7 PBS1 (top) and PCS1 (bottom). Note the faint displaced bundle in the PCS1 zoom — more on it later*

# Quality cut 1: NRMSE ≤ 0.4 — where the number comes from

- NRMSE of fit_ok events is bimodal: good fits (median ≈ 0.05–0.1) vs noise triggers (≈ 1–2), valley at ≈ 0.4–0.5
- The cut sits in the valley — it is read off the distribution, not tuned on the templates
- Weak detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): the PTOF window admits a noise-dominated mixture — this cut is what extracts the real-pulse population



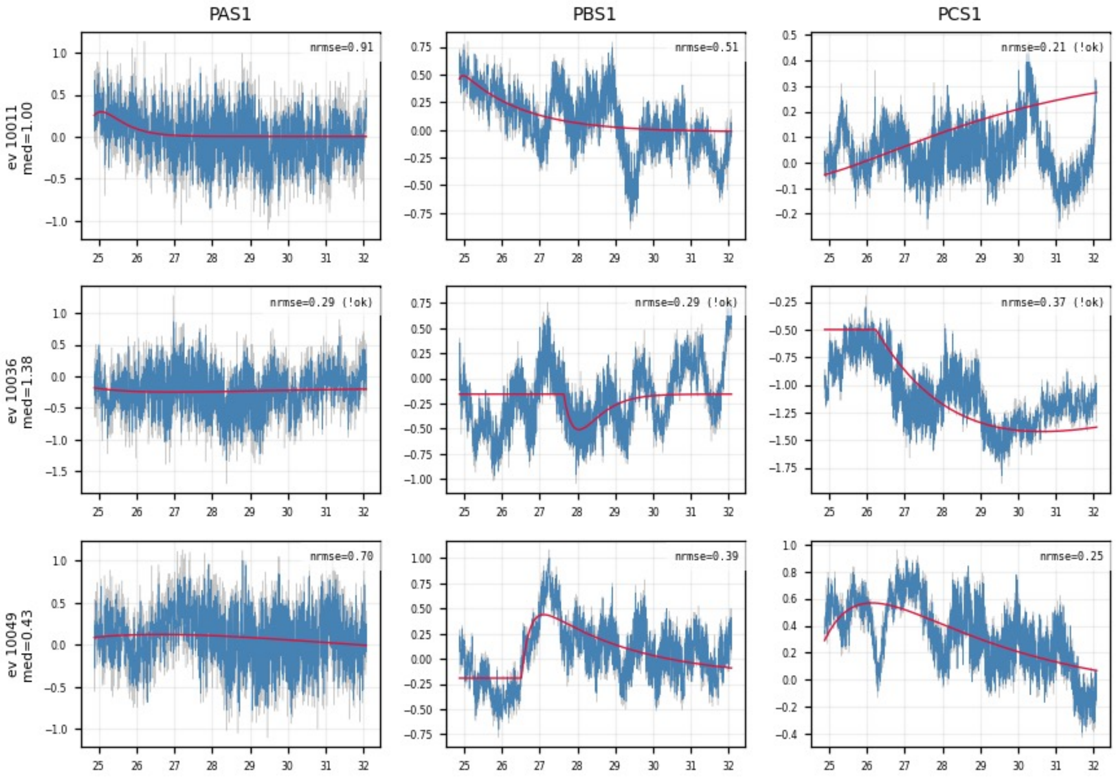

*Z7 PBS1 (quiet): two clean populations, log-log axes*



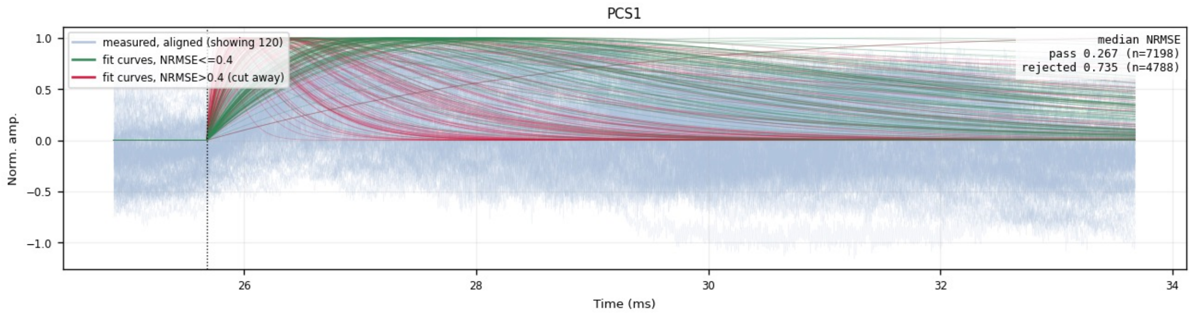

*Z22 PAS1 (weak): noise population dominates — the window alone is not enough*

# The rejected population is noise — verified in the raw traces

- Event grids of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse — the cut removes noise triggers, not physics
- Fan-cut view: fitted curves that pass (green) vs cut away (red) on top of the aligned data, median NRMSE of each population stamped per channel



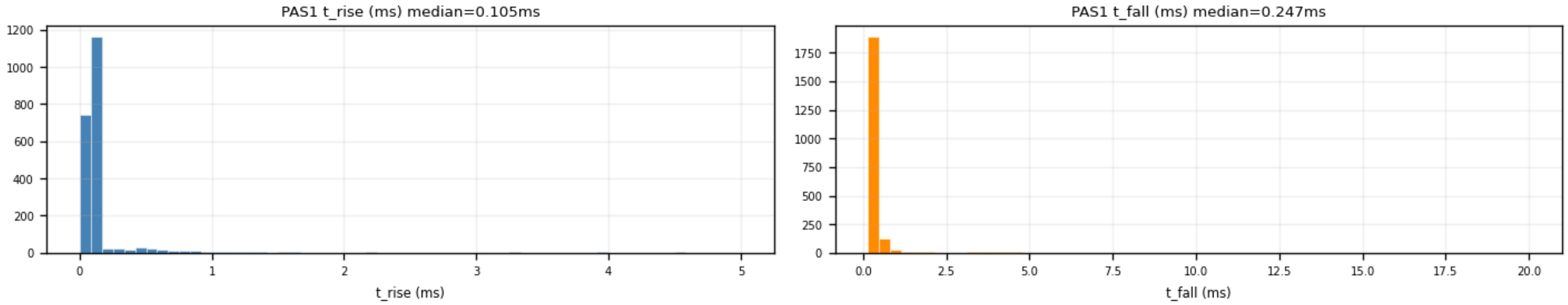

Z22: NRMSE-rejected events — raw traces are noise




Z22 PCS1: pass (green) vs rejected (red)

# Quality cut 2: τ_rise ≤ 0.3 ms — against slow baseline drift

- On noisy-window detectors a smooth, slow baseline drift can survive the NRMSE cut — a slow 2-exp hugs it with a small residual
- The fast-pulse population sits at τ_rise ≈ 0.1 ms (p90 ≈ 0.15 ms) — cleanly separated from the drift tail, so a 0.3 ms ceiling removes the leakage
- Cost ≈ 2–5% of signal on quiet detectors; known trade-off: it also trims the small population of genuine slow-rise pulses (final decision pending)

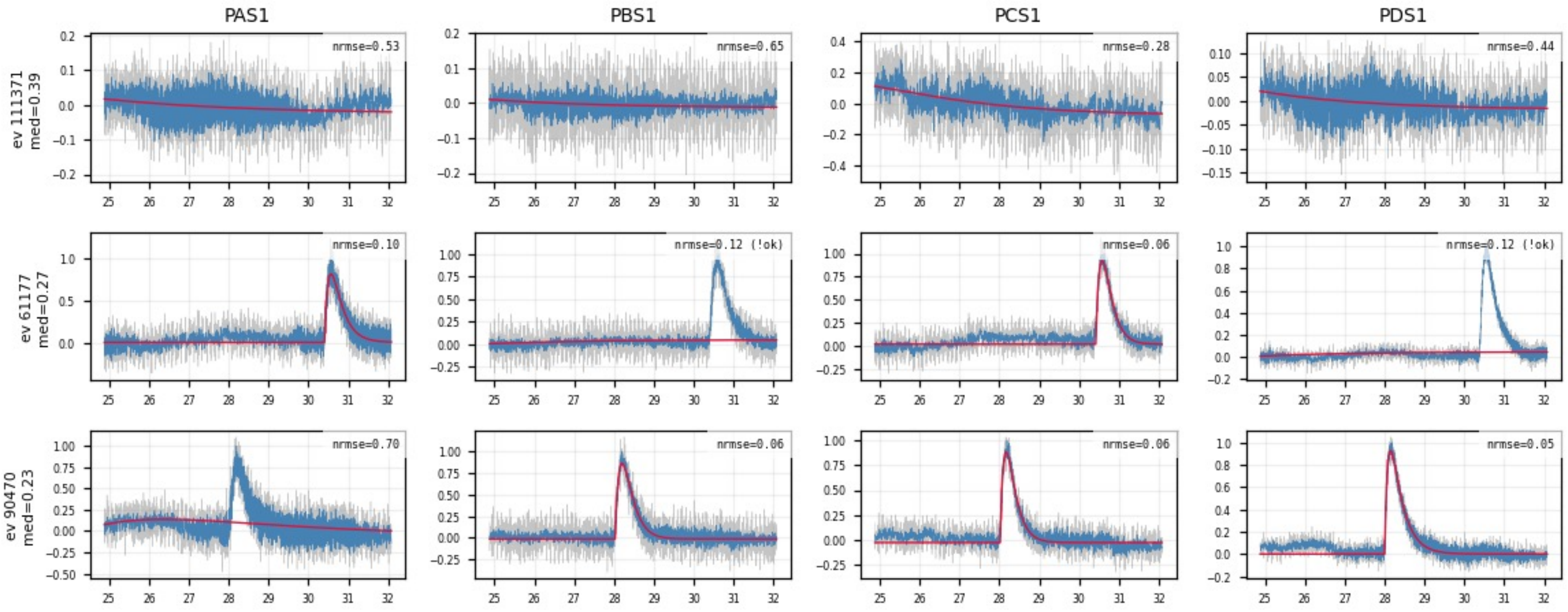



*Z7 PAS1: fitted τ_rise (median 0.105 ms) and τ_fall distributions*

- On quiet detectors both τ_rise and τ_fall distributions are narrow — the pulse shape is stable across events

# A genuine slow-pulse population (“shadow” events)

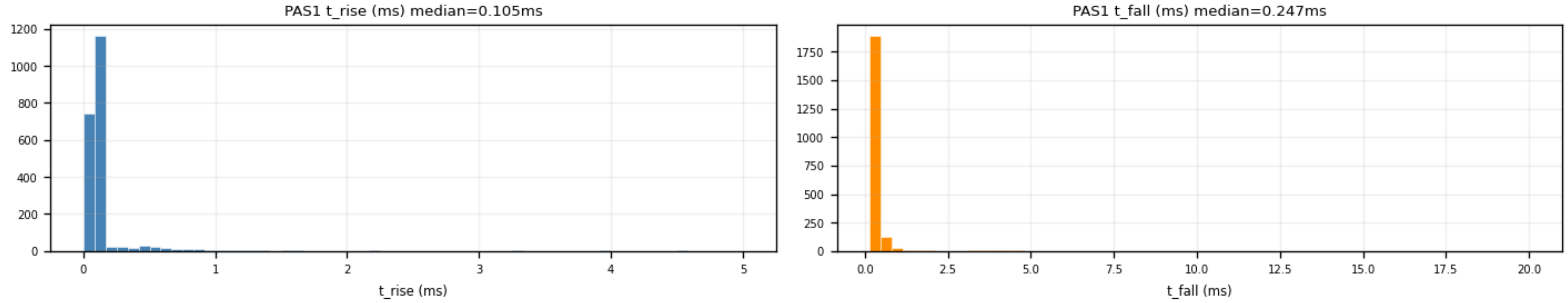
- Selection: well-fit (median NRMSE ≤ 0.4) AND slow (median τ_rise > 0.2 ms) — raw traces show real pulses, not noise
- They are the faint displaced bundle seen in the aligned overlays — real shape variation (candidate surface/bulk effect)
- Exactly the kind of shape variation the multi-template NxM method is built to capture



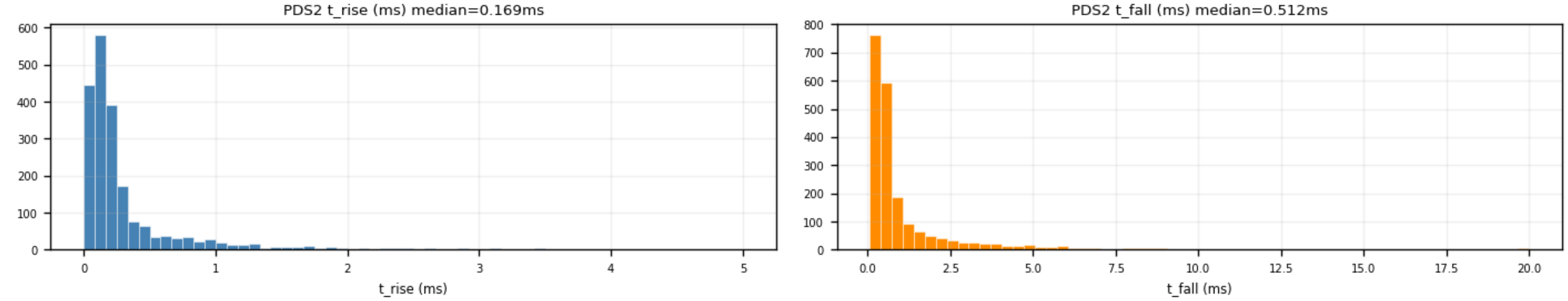

*Z7 shadow events (first 4 channels): raw (gray) / LP (blue) / fit (red) — genuine slow pulses, onset aligned, peak late*

# Side finding: long-τ_fall events on Z7 are a one-channel artifact

- PDS2 shows a broad τ_fall tail (median 0.51 ms vs ≈ 0.25 ms elsewhere) and a smeared τ_rise distribution
- Cross-channel check: the same events are normal fast pulses in the other 11 channels → a PDS2-only low-frequency disturbance, not slow physics



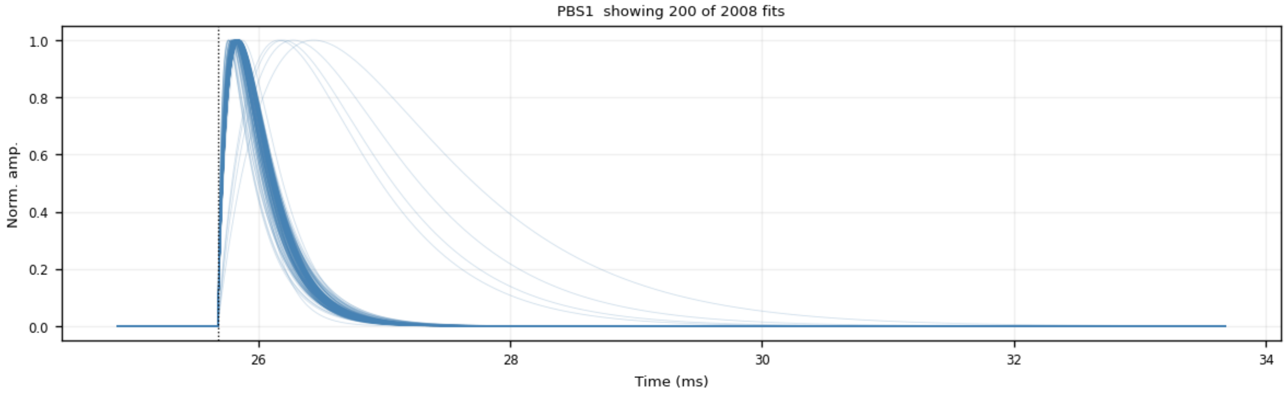

*Z7 PAS1 — reference channel: narrow τ_rise / τ_fall*

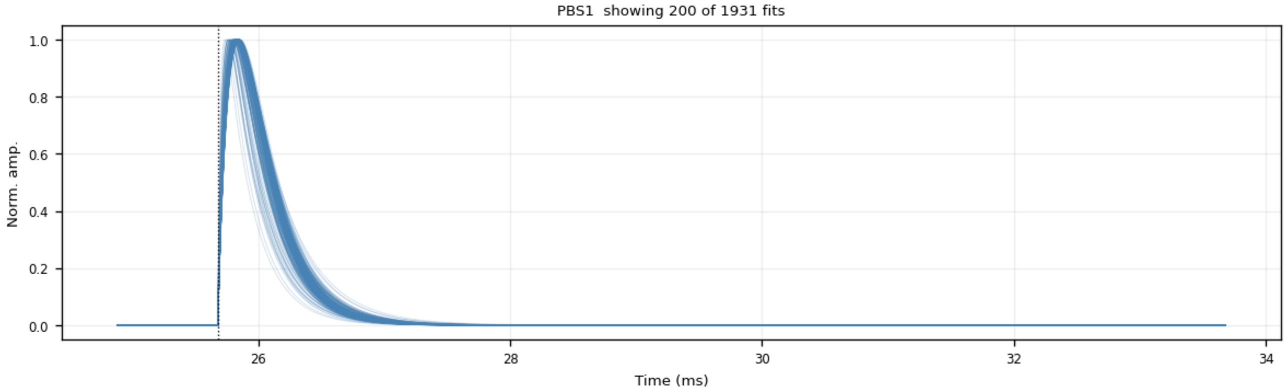



*Z7 PDS2 — broad tails from a channel-specific low-frequency artifact*

# Template family 1 — analytic 2-exp, NRMSE-weighted (1x1)

- Per channel: weighted mean of ALL fit_ok fitted curves at the common onset, weight $w = 1/\max(\mathrm{NRMSE}, 0.01)^2$
- Badly-fit events count less but are never excluded — robust and smooth (noise-free by construction)
- Delivered as peak-normalized 32768-bin ROOT TH1D per channel + summed PT/PS1/PS2 templates

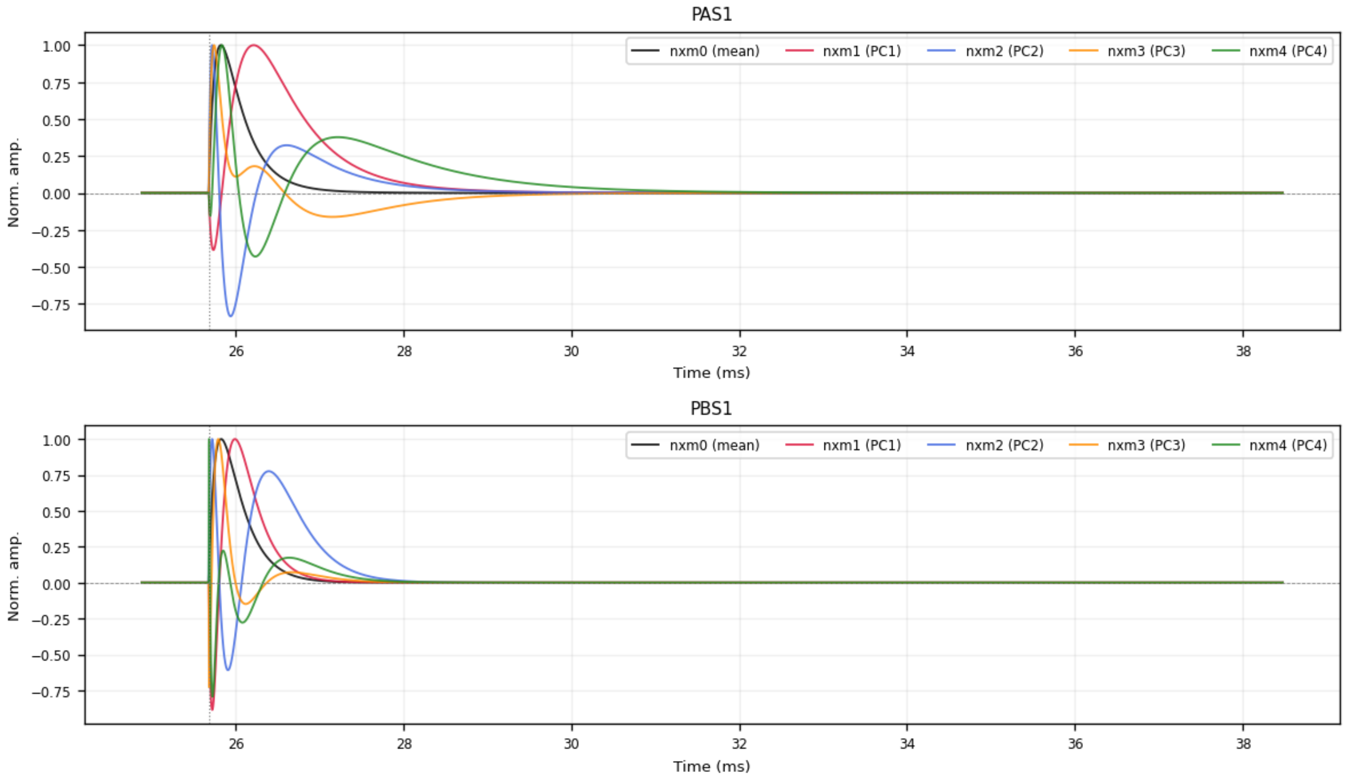



*…after NRMSE ≤ 0.4 and τ_rise ≤ 0.3 ms — a single tight shape family; the weighted mean of these is the 1x1 template*

# Template family 2 — NxM PCA templates

- Input: fitted curves passing fit_ok + NRMSE ≤ 0.4 + $\tau$_rise ≤ 0.3 ms, at common onset, peak-normalized (PCA window 15550–24050, ≤ 3000 curves/channel)
- nxm0 = mean shape;  nxm1…nxm4 = first four principal components — a real pulse is fit as $\Sigma_i\ amp_i \cdot nxm_i$
- PC1 + PC2 already capture 96–98% of the shape variance; all five delivered peak-normalized




*Z7 PAS1 / PBS1: nxm0 (mean) + nxm1–4 (PCs) — the oscillating components encode rise/fall-time variation*

# Delivered — and what remains

- Delivered for all 13 zips, official cdmsbats PulseTemplates layout (zip{N}/{chan}, {chan}nxm0–4, summed PT/PS1/PS2):
  - SNOLAB_R4_20260706_ZhihengLi_zip{N}.root — 2-exp NRMSE-weighted (1x1)
  - SNOLAB_R4_20260707_ZhihengLi_pca_zip{N}.root — NxM PCA (normalized)
- Every step is traceable: raw cache → per-series fit checkpoints indexed by EventNumber → self-documenting figures (11 types × 13 zips, all versioned)
- Cuts derived from the data, validated in raw traces: NRMSE ≤ 0.4 (bimodal valley), $\tau$_rise ≤ 0.3 ms (drift leakage)
- Open items:
  - final decision on the $\tau$_rise ceiling vs the genuine slow-pulse population
  - run the group's template-validation method on the new templates